

PPA Exposure Model

Methodology Document

Prepared for: Credit and Liquidity Risk Team

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1. Overview

This document describes the methodology for a Monte Carlo counterparty exposure model applied to a fixed-for-float Power Purchase Agreement (PPA). The model estimates the potential future credit exposure over the PPA lifetime, under both uncollateralized and collateralized (CSA) scenarios.

The model supports two forward curve modes: (i) a model-implied synthetic curve derived from the Schwartz parameters, and (ii) a market-anchored mode where the user provides an observed forward curve and the simulation generates stochastic deviations around it. The market-anchored mode is the standard approach in production risk systems.

2. Price Model Specification

2.1 Stochastic process

The underlying power price follows a mean-reverting process in log-space (Ornstein-Uhlenbeck). Letting $X(t) = \ln(S(t))$:

$$dX_t = \kappa (\mu(t) - X_t) dt + \sigma dW_t$$

where the seasonal long-run mean is:

$$\mu(t) = \mu_{base} + A \sin(2\pi(t - \phi))$$

Parameter	Symbol	Description	Default
Mean-reversion speed	κ	Speed of reversion to equilibrium	1.5
Long-run mean	μ_{base}	$\ln(\text{equilibrium price})$	$\ln(52)$
Volatility	σ	Annualized log-price volatility	0.35
Seasonality amplitude	A	Seasonal swing in log-price	0.15
Seasonality phase	ϕ	Phase shift (0 = winter peak)	0.0

2.2 Discretization (Euler-Maruyama)

The SDE is discretized with monthly time steps ($\Delta t = 1/12$):

$$X_{t+\Delta t} = X_t + \kappa (\mu(t) - X_t) \Delta t + \sigma \sqrt{\Delta t} Z$$

where $Z \sim N(0,1)$. Prices are recovered as $S(t) = \exp(X(t))$.

2.3 Conditional distribution

The OU process has a known conditional distribution:

$$X_T | X_t \sim \mathcal{N}(m(t, T), v(t, T))$$
$$m(t, T) = e^{-\kappa\tau} X_t + (1 - e^{-\kappa\tau}) \mu(T)$$
$$v(t, T) = \frac{\sigma^2}{2\kappa} (1 - e^{-2\kappa\tau})$$

where $\tau = T - t$.

2.4 Forward price

The model-implied forward price for delivery at T:

$$F(t, T) = \exp \left(m(t, T) + \frac{v(t, T)}{2} \right)$$

This ensures MtM valuation at each time step uses information consistent with the simulated spot price, avoiding look-ahead bias.

2.5 Market-anchored simulation

When a market forward curve is provided, the simulation generates stochastic deviations around it. Let $Y(t)$ be the deviation from the market forward in log-space:

$$dY_t = -\kappa Y_t dt + \sigma dW_t, \quad Y_0 = 0$$

The anchored price is:

$$S_t = F_{mkt}(0, t) \cdot \exp \left(Y_t - \frac{\text{Var}[Y_t]}{2} \right)$$

The variance correction ensures $E[S(t)] = F_{mkt}(0, t)$, guaranteeing that expected simulated prices match observed forwards at every tenor.

3. PPA Valuation

The PPA is a fixed-for-float contract modeled from the generator's perspective.

Mark-to-Market at time t :

$$MtM(t) = \sum_{i: t_i > t} (K - F(t, t_i)) \cdot Q_i \cdot DF(t, t_i)$$

where:

- K = fixed price (EUR/MWh)
- $F(t, t_i)$ = model-implied forward at time t for delivery at t_i
- Q_i = Capacity $\times$ Capacity Factor $\times$ Hours per period (MWh)
- $DF(t, t_i) = \exp(-r \cdot (t_i - t))$ = continuous discount factor

4. Exposure Metrics

$$EE(t) = \mathbb{E}[\max(MtM(t), 0)]$$

$$PFE_{\alpha}(t) = Q_{\alpha}[\max(MtM(t), 0)]$$

$$EPE = \frac{1}{T} \sum_{i=1}^N EE(t_i) \Delta t$$

Metric	Use
EE — Expected Exposure	Average credit exposure at each time point
PFE — Potential Future Exposure	Tail exposure for credit limit setting (97.5% or 99%)
EPE — Expected Positive Exposure	CVA calculation input
Effective EPE	Non-decreasing EPE — regulatory capital (Basel III / SA-CCR)
Peak PFE	Maximum PFE across all horizons — credit limit allocation

5. Collateral Modeling

The model supports CSA collateral with threshold (H), Minimum Transfer Amount (MTA), Independent Amount (IA), and Margin Period of Risk (δ):

$$C(t) = \max(MtM(t - \delta) - H, 0) + IA$$

subject to the MTA filter. The MPOR introduces a lag between MtM observation and collateral receipt. Net exposure:

$$Net\ Exposure(t) = \max(MtM(t) - C(t), 0)$$

6. Model Validation

6.1 Convergence analysis

The model runs simulations with increasing path counts (500 to 50,000) and tracks Peak PFE stability and standard error of mean exposure. Convergence is assessed by the relative difference between successive counts.

6.2 Distribution diagnostics

MtM distribution at each time step is analyzed for skewness, excess kurtosis, and normality (Jarque-Bera test). A QQ plot compares empirical vs normal reference. Commodity-linked exposures typically show fat tails, making PFE at high confidence levels sensitive to distributional assumptions.

7. Assumptions and Limitations

- 1. Single stochastic factor.** One factor drives the entire curve. Multi-factor models would capture term structure dynamics and short/long-term decorrelation.
- 2. No jump component.** Power price spikes are not captured. Jump-diffusion extensions would improve tail estimation.
- 3. Constant volatility.** Volatility assumed constant. Implied vol varies by delivery period and conditions.
- 4. Deterministic generation.** Capacity factor is fixed. Renewable generation is stochastic and price-correlated.
- 5. Simplified collateral.** Does not capture eligible collateral types, rehypothecation, multi-contract netting, or wrong-way risk.

These are documented as areas for future development. The prototype provides a sound foundation for incremental extension.

8. References

- Schwartz, E.S. (1997). *The Stochastic Behavior of Commodity Prices*. Journal of Finance, 52(3).
 - Gregory, J. (2020). *The xVA Challenge*. 4th Ed. Wiley.
 - Burger, M., Graeber, B., & Schindlmayr, G. (2014). *Managing Energy Risk*. 2nd Ed. Wiley.
 - Basel Committee (2014). *SA-CCR: Standardised Approach for Counterparty Credit Risk*.
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